homework02

P89-习题 8.2 (A 类) 1-3, 1-5, 1-7; 3-1; 6;
P99-习题 8.3 (A 类) 1-3, 1-6; 6; 7*; 13**

1-3、求一阶、二阶偏导数 $z = x^y$

$$z_x = y\,x^{y-1} \Rightarrow z_{xx} = y(y-1)\,x^{y-2}, \quad z_{xy} = x^{y-1} + y\,x^{y-1}\,\ln x$$
$$\Rightarrow z_y = x^y \ln x \Rightarrow z_{yx} = y\,x^{y-1}\,\ln x + x^{y-1}, \quad z_{yy} = x^y (\ln x)^2$$

1-5、求一阶、二阶偏导数 $z = \dfrac{x}{\sqrt{x^2+y^2}}$

$$z_x = \frac{1}{\sqrt{x^2+y^2}} - \frac{x}{2}\frac{2x}{(x^2+y^2)^{3/2}} = \frac{y^2}{(x^2+y^2)^{3/2}}$$
$$\Rightarrow z_{xx} = \frac{-3}{2}\frac{y^2\,2x}{(x^2+y^2)^{5/2}} = \frac{-3xy^2}{(x^2+y^2)^{5/2}}$$
$$\Rightarrow z_{xy} = \frac{2y}{(x^2+y^2)^{3/2}} - \frac{3}{2}\frac{y^2\,2y}{(x^2+y^2)^{5/2}} = \frac{2yx^2 - y^3}{(x^2+y^2)^{5/2}}$$
$$z_y = \frac{-1}{2}\frac{x\,2y}{(x^2+y^2)^{3/2}} = \frac{-xy}{(x^2+y^2)^{3/2}}$$
$$\Rightarrow z_{yx} = \frac{-y}{(x^2+y^2)^{3/2}} + \frac{3}{2}\frac{xy\,2x}{(x^2+y^2)^{5/2}} = \frac{2x^2y - y^3}{(x^2+y^2)^{5/2}}$$
$$z_{yy} = \frac{-x}{(x^2+y^2)^{3/2}} + \frac{3}{2}\frac{xy\,2y}{(x^2+y^2)^{5/2}} = \frac{2\,xy^2 - x^3}{(x^2+y^2)^{5/2}}$$

1-7、求一阶、二阶偏导数 $z = \arctan \frac{x}{y}$

$$z_x = \frac{1}{1+(\frac{x}{y})^2}\frac{1}{y} = \frac{y}{x^2+y^2} \Rightarrow z_{xx} = \frac{-2xy}{(x^2+y^2)^2}$$

$$\Rightarrow z_{xy} = \frac{1}{x^2+y^2} - \frac{y\,2y}{(x^2+y^2)^2} = \frac{x^2-y^2}{(x^2+y^2)^2}$$

$$\Rightarrow z_y = \frac{1}{1+(\frac{x}{y})^2}\frac{-x}{y^2} = \frac{-x}{x^2+y^2} \Rightarrow z_{yy} = \frac{2xy}{(x^2+y^2)^2}$$

$$\Rightarrow z_{yx} = \frac{-1}{x^2+y^2} + \frac{x\,2x}{(x^2+y^2)^2} = \frac{x^2-y^2}{(x^2+y^2)^2}.$$

3-1、验证等式: $z=\dfrac{x-y}{x+y}\ln\dfrac{y}{x}$, $x\dfrac{\partial z}{x}+y\dfrac{\partial z}{y}=0$

解: 依题意

$$z_x=\Big(1-\frac{2y}{x+y}\Big)'\ln\frac{y}{x}+\frac{x-y}{x+y}\frac{\partial}{\partial x}\big(\ln\frac{y}{x}\big)=\frac{2y}{(x+y)^2}\ln\frac{y}{x}-\frac{x-y}{x+y}\frac{1}{x}$$

$$z_y=\Big(\frac{2x}{x+y}-1\Big)'\ln\frac{y}{x}+\frac{x-y}{x+y}\frac{\partial}{\partial y}\big(\ln\frac{y}{x}\big)=\frac{-2x}{(x+y)^2}\ln\frac{y}{x}+\frac{x-y}{x+y}\frac{1}{y}$$

$$\Rightarrow x\frac{\partial z}{x}+y\frac{\partial z}{y}=\frac{2xy-2xy}{(x+y)^2}\ln\frac{y}{x}+\Big(-\frac{x-y}{x+y}+\frac{x-y}{x+y}\Big)=0$$

6、已知 $z=z(x,y)$ 由 $z=\displaystyle\int_y^{xy}y\,e^{t^2}\,dt$ 表示, 求 $\dfrac{\partial z}{\partial x}$

解: 记 $F(x)=\displaystyle\int_0^x e^{t^2}\,dt$, 则 $F(0)=0, F'(t)=e^{t^2}$

$$z=y\big(F(xy)-F(y)\big)\Rightarrow z_x=y\,F'(xy)\,y=y^2e^{(xy)^2}.$$

1-3、$u = z^2 + y^2 + yz,\ z = \sin t,\ y = e^t$, 求 $\dfrac{du}{dt}$

解: 依题意

$$\frac{du}{dt} = u_z z'(t) + u_y y'(t) = (2z + y)\cos t + (2y + z)\, e^t$$

1-6、$z = f(x^2 - y^2, e^{xy})$, 求 z_x, z_y

$$z_x = f_1\, 2x + f_2\, e^{xy}\, y = 2x\, f_1 + y\, e^{xy}\, f_2$$
$$z_y = f_1(-2y) + f_2\, e^{xy}\, x = -2y\, f_1 + x\, e^{xy}\, f_2$$

6、设 $f(x,y)=g(r), r=\sqrt{x^2+y^2}$, 求证

$$\frac{\partial^2 f}{\partial x^2}+\frac{\partial^2 f}{\partial y^2}=g''(r)+\frac{1}{r}g'(r).$$

证: 依题意, $r_x=x/r, r_y=y/r$,

$$\Rightarrow f_x=g'(r)r_x=g'(r)\frac{x}{r}$$

$$\Rightarrow f_{xx}=g'(r)r_x\frac{x}{r}+g'(r)\Big(\frac{1}{r}-\frac{x}{r^2}r_x\Big)=g''(r)\frac{x^2}{r^2}+g'(r)\frac{y^2}{r^3}$$

$$f_{yy}=g''\frac{y^2}{r^2}+g'(r)\frac{y^2}{r^3}\Rightarrow f_{xx}+f_{yy}=g''(r)+\frac{1}{r}g'(r)$$

7*、设 $u(x,y)$ 的所有二阶偏导数都连续, 且 $u_{xx}-u_{yy}=0$, $u(x,2x)=x$, $u'_x(x,2x)=x^2$, 试求 $u_{xx}(x,2x)$, $u_{xy}(x,2x)$, $u_{yy}(x,2x)$.

解: 依题意, 记 $f(x)=u(x,2x)$, 则

$$f(x)=x \quad f'(x)=u_1+2\,u_2=1 \Rightarrow u_2(x,2x)=\frac{1-x^2}{2}$$

$$\Rightarrow u_{21}+2u_{22}=-x, \quad u_{11}+2u_{12}=2x, \quad u_{11}-u_{22}=0$$

$$\Rightarrow u_{11}=-\frac{4}{3}x, \quad u_{12}=\frac{5}{3}x, \quad u_{22}=\frac{-4}{3}x$$

13** 设 $z = z(u, v)$ 且变换 $\begin{cases} u = x - 2y \\ v = x + ay \end{cases}$，可把方程 $6z_{xx} + z_{xy} - z_{yy} = 0$，简化为 $z_{uv} = 0$，求 a.

解：依题意，

$$z_x = z_u u_x + z_v v_x = z_u + z_v$$

$$z_{xx} = (z_{uu} u_x + z_{uv} v_x) + (z_{vu} u_x + z_{vv} v_x) = z_{uu} + 2z_{uv} + z_{vv}$$

$$z_{xy} = z_{uu} u_y + z_{uv} v_y + z_{vu} u_y + z_{vv} v_y = -2z_{uu} + a\, z_{uv} + -2z_{vu} + a\, z_{vv}$$

$$z_y = z_u u_y + z_v v_y = -2\, z_u + a\, z_v$$

$$z_{yy} = -2\left(z_{uu} u_y + z_{uv} v_y\right) + a\left(z_{vu} u_y + z_{vv} v_y\right) = 4\, z_{uu} - 4a\, z_{uv} + a^2\, z_{vv}$$

$$\Rightarrow 6\, z_{xx} + z_{xy} - z_{yy} = z_{uu}\,(6 - 6) + z_{uv}\,(10 + a + 4a) + z_{vv}\,(6 + a - a^2)$$

$$\Rightarrow 6 + a - a^2 = 0, \quad 10 + 5a \neq 0 \Rightarrow a = 3.$$